

The WTO Dispute Dilemma: Understanding the Extent Third-Parties Complicate Compliance

Sophia Lander and Jessica Gallardo

Instructor: Allison Craig

Class: Statistical Analysis in Political Science II

June 22, 2026

Abstract

To what extent do third-party observers truly decrease compliance rates of respondent countries in WTO litigation? This paper replicates and extends Kucik & Peritz (2021), who argue that greater third-party participation decreases respondents' compliance with WTO rulings. We confirm that their core findings are robust and address two important limitations of the paper's argument. First, leveraging splines and binning, we analyze the effect that different levels of third-party participation have on respondent compliance. Second, we control for fiscal capacity, a confounder often cited in the broader literature and acknowledged by Kucik & Peritz (2021), but left uncontrolled in their respective models. We find that the negative effect of third-party participation on compliance may be an artifact of the limited number of cases and skewed distribution of the number of third party participants in any given case. Furthermore, the additional fiscal capacity control does not substantially alter our findings, but marginally improves our overall models. Lastly, our extension reveals that the effect of varied levels of third-party participation on compliance is potentially more nuanced and conditional than the broader literature suggests.

1 Introduction

Does the presence of third party countries increase compliance with World Trade Organization (WTO) disputes rulings—and if so, does that effect depend on how many are involved? According to Kucik and Peritz (2021), third party observers may actually decrease compliance with rulings by the respondent state. Specifically, they find that bystander states, while they may increase the likelihood of a ruling, in the first place, decrease the likelihood of compliance with such a ruling. While their argument runs counter to previous literature, they provide several theoretical reasons as to why compliance might be harder to achieve with third party countries.

The WTO dispute settlement mechanism is one of the most widely used avenues by which states can bring their economic grievances with other states to an adjudicating body. The WTO can then decide, in those cases, to issue a ruling either in favor of the claimant or the respondent state. One way that the WTO tries to increase compliance with the rulings it puts out is by allowing third party participation in the case. Busch and Reinhardt (2006) argue that third party participation in WTO disputes lowers the likelihood that the case settles early—an effect that they say prevents the litigant from pursuing discriminatory settlements. Similar, Maggi (1999) finds that third parties may increase the costs of punishment for the respondent if they do not comply with the ruling.

Kucik and Peritz argue that despite these findings, third parties may actually hinder compliance. One rationale they provide is that third parties bring their own arguments, opinions, and voice to the dispute, which may increase the scope of the ruling and in turn make compliance harder for the respondent state. They back their findings up with the canonical finding from Downs et al. (1996) that stricter enforcement mechanisms in the international arena may backfire. Essentially, though, Kucik and Peritz’s argument boils down to a trade-off between increasing the number of eyes on the dispute and simultaneously increasing the number of voices; even if third parties create pressure for a state to comply, they could also make any such compliance more difficult to achieve, even when the respondent

state would like to comply—leading to an increase in rulings, but decrease in compliance with them.

To investigate their theory, Kucik and Peritz generate a dataset of WTO dispute initiated from 1995 to 2012. Within that, their sample size is 155 disputes, or the ones that ended in a ruling that found the respondent in violation. Of that, only about half of the disputes had a third party present and the remaining had fewer than six involved. Yet, some had up to 24 third parties involved. In this paper, we replicate their study and confirm their central finding: third-party participation is negatively associated with compliance. We then extend their research design in two ways. First, given the heavily skewed distribution of third-party counts, we re-estimate their models using regression splines and binned specifications to test whether the compliance effect is non-linear — specifically, whether small numbers of third parties have minimal impact but large numbers cross a threshold beyond which compliance deteriorates sharply. Second, we introduce a state capacity control to test whether a respondent’s ability to comply moderates the third-party effect. Together, these extensions suggest the relationship Kucik and Peritz identify is more conditional than their logarithmic specification implies.

2 Replication

To build on the work of Kucik and Peritz’s (2021), we replicated their study to understand their empirical decisions and verify their findings. Consistent with their published results, we find that compliance among WTO respondents declines as the number of third-party participants increases. In this study, the authors employ a series of robustness checks, including multiple response-variable measures and statistical models, to ensure generalizability of the results. Furthermore, clustered standard errors were accounted for at the dispute-pair level, and later Heckman Selection models were used to control for nonrandom selection bias into WTO rulings. Finally, it should be noted that we focused solely on replicating the empirical

results of this paper, not recreating the original data set itself.

This study uses a total of eight statistical models, including multiple probits, linear probability models, and Heckman Selection models. Across models, the dependent variable—the natural log of third parties—remained consistent, indicating that Kucik and Peritz’s (2021) findings are not limited to a single model (see Table 2). In addition, these models were tested using different measures of the dependent variable to capture varying levels of compliance. The authors defined compliance in two ways: a dichotomous measure indicating whether the defendant complied or not, and an ordinal measure assessing different levels of compliance. The ordinal measure takes on three degrees of compliance: 0 (no compliance with the WTO ruling), 1 (partial compliance) and 2 (full compliance). Lastly, this study also control for various factors: economic, geopolitical, and WTO-specific procedures.

Economic share controls include the defendant’s Gross Domestic Product (GDP) in proportion to global GDP (*gdp_share*) and the relative share of the product market of the defendants, proportional to the share of the complainants (*trade_share_rctp*). Other controls are more adapted to account for the procedural intricacies of the WTO, such as how many times a defendant has been previously found guilty (*times_ruled*), the logged value of legal claims in the dispute (*ln_claims*) or whether the dispute was considered under Article XXII of the WTO or not (*article_xxii*). Article 22 of the WTO encourages multilateral consultation where third parties can join. The authors controlled for defendants being a geopolitical power like the United States or the EU, 1 if yes, 0 if not (*great_power*). Lastly, binary controls, *remedy* and *clarity*, were controlled for whether the judicial body mandated a specific remedy and whether the ruling was clear.

Prior to replicating the statistical models, we also conducted further descriptive analysis to understand the data and empirical decisions made. Although the total adverse rulings is 155, the United States and the European Union, collectively account for more than half (55%) of the total respondents (see Figure 1). Furthermore, it seems that some many countries were simply in more disputes with the United States and EU than others. Given

the limited sample size, it is imperative to correct for clustered standard errors arising from repeated dispute pairs (or litigant dyads). Table 1 illustrates the case disparity between high-to-low income countries. For example, the EU and the United States appear together as many as 20 times before the WTO judicial body. However, an emerging economy such as Argentina stood in front of the WTO dispute settlement body eight times. We stress this within-dyad correlation in the replication section because it clearly violates the Gauss-Markov assumptions of standard regression: that each individual observation is independent of the others, further supporting Kucik and Peritz’s choice to cluster standard errors for dispute pairs.

Kucik and Peritz demonstrate the validity of their results using multiple models, testing them with different compliance measures, and adding different types of controls throughout their article. As shown in Table 3, across all eight models, there is a significant negative association between third party participation and compliance. Models 1 through 4 confirm robust results for different compliance measures and model functional form (probit vs. linear). Models 5 and 6 controlled for other additional product specific economic factors, such as industry employment share and the log of disputed bilateral trade shares between complainants and defendants; however, both of these controls deemed statistically insignificant (see Table 2). Lastly, Kucik and Peritz pay particular attention to the self-selection bias of third-party participation and whether it drives cases to a formal ruling instead of early settlement. Heckman Selection models 7 and 8 account for additional variables, including systemic interest (*systemic_new*) in the case and the number of times a defendant lost a formal ruling related to the same topic (*adverse_frequency*). Ultimately, both Heckman models find that more third-party participation leads to cases moving into formal rulings, along with more third-party participation leading to a decrease in compliance even after accounting for self-selection. Although these results appear significant, we noticed that the authors may be missing a few key variables that might influence a defendant’s capability to comply with the formal outcome.

Table 1: Repeated Dispute Dyads and Compliance in WTO Adverse Rulings

Country A	Country B	Total Disputes	N Comply	N Non-Comply	Compliance Rate
EU	USA	20	10.00	10	0.50
Canada	USA	10	9.00	1	0.90
Korea	USA	8	8.00	0	1.00
Mexico	USA	8	7.00	1	0.88
China	USA	7	4.00	3	0.57
India	USA	7	7.00	0	1.00
Japan	USA	7	4.00	3	0.57
Brazil	EU	5	3.00	2	0.60
EU	Korea	5	5.00	0	1.00
Argentina	EU	4	3.00	1	0.75
Brazil	USA	4	4.00	0	1.00
Canada	EU	4	2.00	2	0.50
China	EU	4	4.00	0	1.00
EU	India	4	2.00	2	0.50
Brazil	Canada	3	2.00	1	0.67
Canada	Japan	3	3.00	0	1.00
Argentina	Chile	2	1.00	1	0.50
Argentina	USA	2	2.00	0	1.00
Australia	EU	2	0.00	2	0.00
Australia	USA	2	1.00	1	0.50
Chile	EU	2	2.00	0	1.00
EU	Japan	2	2.00	0	1.00
EU	Thailand	2	0.00	2	0.00
Guatemala	Mexico	2	2.00	0	1.00
Indonesia	Japan	2	2.00	0	1.00
Indonesia	USA	2	1.00	1	0.50
New Zealand	USA	2	2.00	0	1.00
Thailand	USA	2	2.00	0	1.00

Table 2: Replication Results

	Compliance						Heckman (7)		Heckman (8)	
	Probit (1)	Ordered (2)	Linear (3)	Linear (4)	Probit (5)	Probit (6)	Selection (7)	Outcome	Selection (8)	Outcome
lnthird	−0.648** (0.194)	−0.562** (0.184)	−0.184** (0.051)	−0.343** (0.105)	−0.763** (0.286)	−0.785* (0.365)	0.634** (0.140)	−0.167* (0.066)	0.604** (0.137)	−0.151* (0.061)
trade_share_rctp	1.270* (0.565)	1.114* (0.567)	0.396* (0.162)	0.708* (0.335)	1.619* (0.691)	2.129* (1.056)	2.177** (0.412)	0.480* (0.201)	2.216** (0.390)	0.530** (0.182)
times_ruled	0.006 (0.008)	0.005 (0.006)	0.002 (0.002)	0.003 (0.004)	−0.001 (0.008)	−0.016 (0.011)	−0.003 (0.005)	0.002 (0.002)	0.002 (0.006)	0.002 (0.002)
gdp_share	−0.689* (0.399)	−0.520 (0.386)	−0.139 (0.109)	−0.284 (0.228)	−1.017* (0.432)	−0.735 (1.629)	−0.202 (0.270)	−0.111 (0.110)	−0.060 (0.264)	−0.118 (0.111)
r_greatpower	0.450 (0.353)	0.377 (0.308)	0.100 (0.104)	0.210 (0.206)	0.990* (0.467)		0.461* (0.224)	0.125 (0.094)	0.506* (0.227)	0.132 (0.094)
article_xxii	0.638* (0.278)	0.535* (0.248)	0.172* (0.080)	0.334* (0.160)	0.466 (0.415)	0.751 (0.689)	−0.277 (0.198)	0.117 (0.077)	−0.211 (0.195)	0.106 (0.078)
remedy	−0.029 (0.306)	0.083 (0.286)	0.008 (0.086)	0.055 (0.177)	0.037 (0.440)	0.100 (0.904)	−0.204 (0.212)	0.069 (0.072)	−0.234 (0.186)	0.068 (0.072)
clarity	−0.795* (0.317)	−0.545* (0.308)	−0.217* (0.093)	−0.362* (0.192)	−0.784* (0.448)	−1.697 (1.037)				
systemic_new							0.764** (0.231)		0.696** (0.226)	
adv_frequency									−0.646** (0.246)	
Constant	0.895 (0.634)		0.738** (0.171)	1.349** (0.358)	0.753 (0.654)	0.904 (2.096)	−1.634** (0.309)	0.528 (0.270)	−1.778** (0.442)	0.452* (0.229)
Observations	153	153	153	153	115	61	373	373	373	373
R^2			0.219	0.190						
Adjusted R^2			0.175	0.145						

Note: Standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Frequency of WTO Adverse Rulings by Defendant Country (1995–2012)
155 Total Adverse Rulings

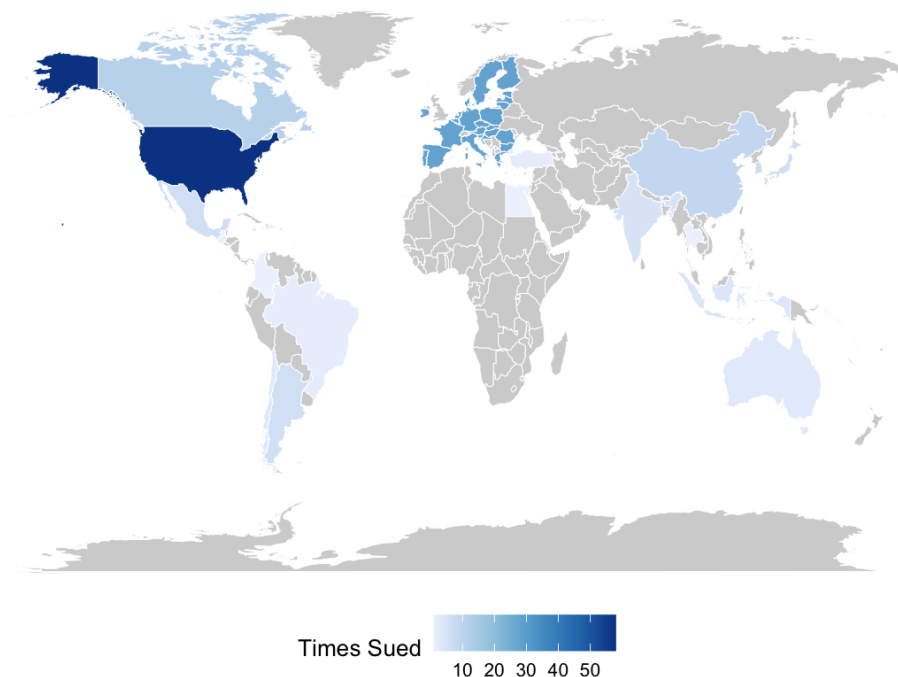


Figure 1: Defendant Country Frequency in WTO Adverse Rulings

3 Amendment and Findings

Figure 1 illustrates the distribution of the number of third parties in cases which ended in a ruling. As Kucik and Peritz (2021) report, there is a heavy left-skewed-ness to the independent variable. In order to account for this skewed-ness the authors use the natural log of the variable. While this is a valid statistical approach to reduce heteroskedasticity and other OLS assumption violations, the process of fitting the entire dataset to the linear model by design, generalizes the findings across all levels of third party participation. For instance, the replicated findings in Table 1 (Model 3) indicates that a one-unit increase in the number of third parties results in a -0.184 percentage point change in likelihood that that case’s ruling was complied with.

The problem however with this interpretation of their findings is two fold. First, many cases never had any third parties to begin with. Yet, the way that (Kucik and Peritz, 2021)’s model is setup, going from no third parties to one third party is treated the same

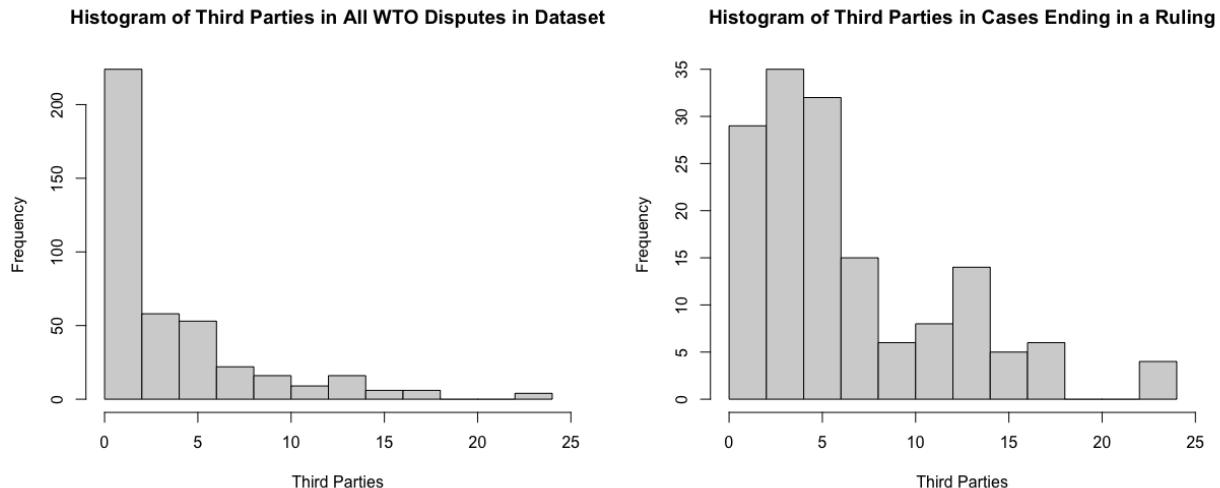


Figure 2: Distribution of Third Party Participants in WTO Cases

as going from five to six third parties. With nearly half of the cases in the dataset having no third parties Kucik and Peritz (2021) (and 5% of cases that reached a ruling), there may be reasonable skepticism about whether the first-unit change is more than just another increment change.

Second, the theoretical argument that the authors put forward is that as the number of third parties increases, the greater chance there is for third parties to essentially "muddy the water" of the ruling by putting forward their own demands for the ruling—making it increasingly difficult for the respondent state to comply. This is in contrast with the existing prior literature that argued that increasing the number of 'eyes' on the ruling, would increase compliance because of reputational and marginal costs of non-compliance increase (Maggi, 1999; Busch and Reinhardt, 2006). The logical extreme of both these arguments is that the greater number of third parties there are in a case, the greater the effect on compliance there should be.

But, at great enough numbers of third parties, what is the substantial difference between 10 and 11 third parties? If third parties obscure rulings and make compliance more difficult, than those effects likely don't occur at low numbers; a one-unit increase at large counts matters far more than one added at small counts. This would suggest that the relationship

between third party participation and compliance is inadequately captured by a linear or logarithmic model. Explicit, neither model captures the marginal effects of each additional third party. Chayes and Chayes (1993) argue that non-compliance stems not from deliberate cost-benefit analysis but from ambiguity. If ambiguity is what drives non-compliance, then the ambiguity-generating effect of third parties likely kicks-in quickly with only a few participants.

For both of the reasons presented, we offer two extensions to Kucick and Peritz’s model. First, we re-estimate both the probit and linear models using regression splines. Second, we estimate the model using binned specifications as a robustness check. Regression splines can account for complex, non-linear functional forms, by dividing the data into segments and then fitting each separately to discover local patterns within the data. By using them, we should be able to tell how the effects of third party participation on compliance varies with the number of third parties—allowing us to address the possibility of limited effects of third party participation in small numbers. By binning the data as a robustness check to the splines, we can both confirm the findings from using the splines and also gauge the optimal levels of third party participation. Both methods allow us to then make predictive estimates of the effect of third party participation and address the following two hypotheses:

Hypothesis 1. *Third party participation has a minimal effect on compliance with WTO rulings when few third parties are involved, and this effect strengthens as the number of third parties increases.*

Hypothesis 2. *There is a substantial lack of discontinuity between having no third-party and having at least one involved, implying that there is not some discrete threshold compliance behavior at no third parties.*

Lastly, given that Kucik and Peritz (2021) and other’s argument about compliance is largely an argument about state capacity or ability to comply to rulings, we introduce another variable the model in order to control for state capacity. This gives us our final hypothesis which connects to H1 and H2 directly:

Hypothesis 3. *The effects of third party participation on compliance will be a factor of state capacity to be able to comply with rulings.*

Implementing regression splines is relatively simple. Without changing any of the models substantively from Kucik and Peritz (2021) we implement natural splines on the dependent variable (natural log of third parties). To interpret the results of the splines, we then calculate the marginal predicted compliance estimate from each of the spline models and then plot them to visualize how compliance changes with increases in the number of third parties. Figure 3 and shows the predicted estimate of compliance (binary on the left panel and ordinal measure on the right) against the natural logarithm of third parties. The left hand panel shows the marginal effects of third parties on compliance for the binary measure of compliance. Noticeably the effect of third parties on compliance does seem to increase with the number of third parties, but the relationship is heavily non-linear. With around no third parties to about seven ($\ln(x) = 2$), the estimated compliance is close to 1 and gradually declining. After about seven third parties, there is then a steep inflection point and the predicted levels of compliance dramatically decreases. With around 20 or more third parties, predicted estimate of compliance drops to around 50%, suggesting that a threshold by which the number of third party observers has an effect on compliance with WTO disputes.

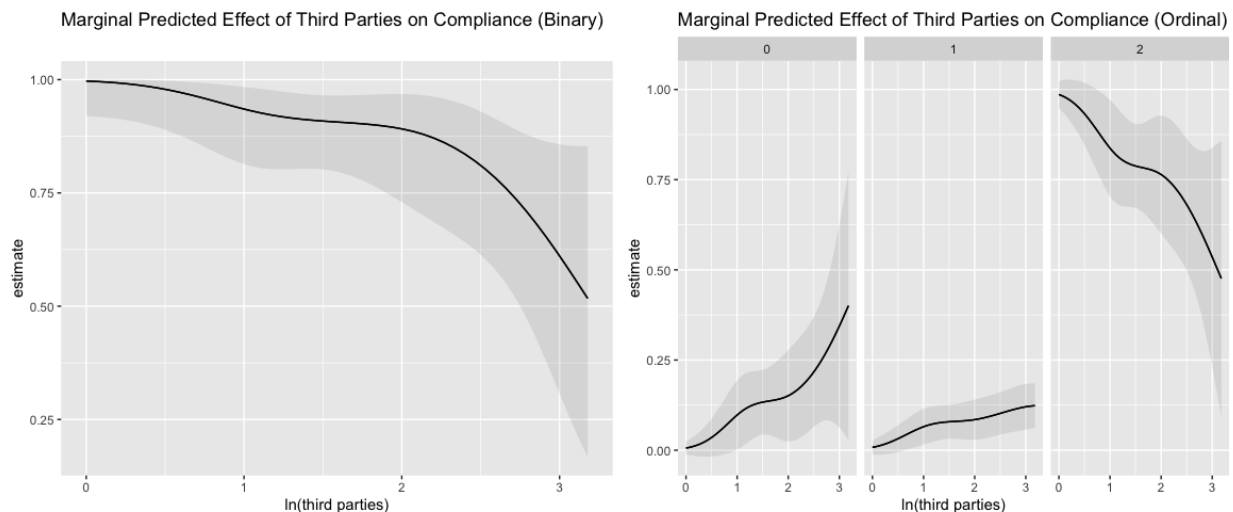


Figure 3: Re-estimation of Probit Models 1 and 2 using Splines

The right hand panel demonstrates the marginal predicted effect of third parties on compliance for the ordinal measure of compliance broken down into the three levels of compliance Kucik and Peritz (2021) describe: 0 = non-compliance, 1 = moderate compliance, 2 = full compliance. For rulings where respondent states did not comply (group 0), as third parties increases the predicted probability of non-compliance rises gradually from about 0 to 0.40. Confirming the results in the left panel, when there are about 3 or less third parties the effect is minimal. Similarly, for rulings where respondent states moderately complied (group 1) the effect of third parties on the predicted probability of compliance is relatively flat and constant. Lastly, for rulings where respondent states did comply, as third parties increase the predicted probability of compliance falls steeply. However, it should be noted that the confidence intervals between group 0 and 2 are quite wide, especially at higher values of third parties, which is likely do to a limited number of observations with higher numbers of third parties.

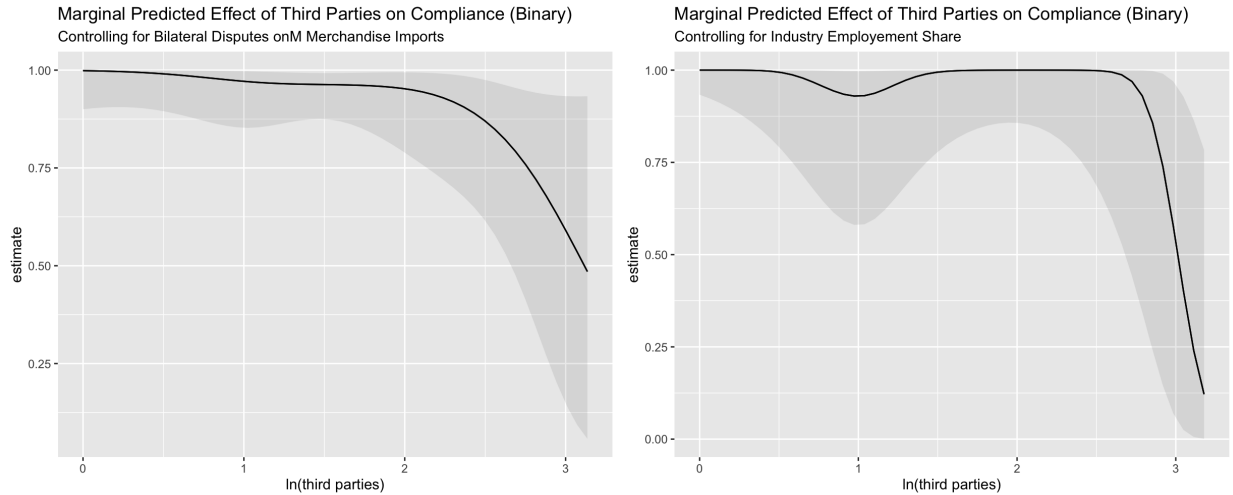


Figure 4: Re-estimation of Probit Models 5 and 6 using Splines

The plots from Figure 4 illustrate the marginal effects of third parties on compliance in the re-estimated Probit 5 and 6 models. These models, as they do in Kucik and Peritz (2021) introduce new control variables. Both panels show a negative relationship between the number of third parties and predicted estimated compliance, with a sharp decline after

about seven third parties, involved. However, the right panel, which introduced a control for industry employment share has an interesting bump in the predicted estimated compliance centered around $\ln(x) = 1$. Additionally, the confidence interval on the right-hand side of this panel is extremely wide, spanning nearly the whole range of the y-axis. It is possible that industry employment share might be correlated with the number of third parties causing a collinearity problem. Or the splines could have resulted in less observations in that knot of data than in others. Either way, the introduction of this control variable clearly makes the model less reliable than the other ones. This isn't to say that this control variable is not substantively interesting, but further investigation into it's concerning behavior in the model is outside of the scope of this paper.

Consider the re-estimates of Models 1, 2 and 5. Notice that as the number of third parties increases in both models, the confidence intervals widen, indicating greater model uncertainty in high numbers of third parties. This is consistent with what we should expect from the data, since there are very few instances of disputes involving more than six third parties, as Kucik and Peritz (2021) report (see Figure 2). Yet, what this tells us is that without further investigation, there is some concern that the effect in question, namely a substantial decrease in the estimate of compliance after seven third parties, might just be an artifact of a few observations and not a statistically meaningful, effect. This is actually where our second model extension, binning, can be a powerful tool.

Binning is a technique by which we convert the continuous independent variable, third parties, into binary, discrete, categories or bins. We then re-estimate the same models from before for each bin, rather than fitting the model to all the entire independent variable. Yet the problem with binning is that it requires the researcher to make arbitrary distinctions about where the bins should be, and how large they should be. However, from the plots in Figure 3, it may be easier to justify the cutoffs for each bin.

Ideally, we'd like to create a bin for no third parties to just one third party to test hypothesis 2. However, when we run Model 1 and 5 on this bin, it creates separation,

i.e., the predictor variable perfectly predicts the binary outcome. This could be due to a number of factors. However, it is likely because within this bin there are a total of 14 case observations, only one of which resulted in non-compliance. So separation is likely occurring because the model in the limited bin only has one data point to learn compliance from. There is no statistical test to overcome this limitation, but it is interesting to note that 93% of cases with no third party or just one third party resulted in compliance. Although we cannot draw the same kind of standard errors or confidence in this finding. Dropping the limited bin, we then create three bins representing, low (roughly 0 to 2 third parties), medium (roughly 3 to 7 third parties), and high (greater than 7) numbers of third parties. Table 3 indicates the bins and the binary compliance outcome for the cases in each bin.

	low	medium	high
0 - non compliance	4	19	17
1 - compliance	25	57	32

Table 3: Compliance Classified by Bin including Limited Bin

We re-estimate Models 1, 2, and 5 on the low bin. Model 1 results in separation again. Analysis on this separation indicates that another variable is creating this result: the control variable for great powers. Essentially, what this means is that all the cases in which non-compliance occurs in the low bin (4 cases) involved great powers, such that great powers perfectly or quasi perfectly predicts the outcome variable. This is an interesting result in and of itself, but in order to estimate Model 1 we drop the great power variable and separation is no longer found in the low bin. We feel this is a valid approach, especially given that Model 2, which uses the ordinal measure of compliance is not susceptible to separation like the binary measure, such that we can estimate the effects of third parties in the low bin with all the same control variables as in Kucik and Peritz (2021). We re-estimate Models 1 and 2 for the medium and high bins as well. Model 5, which added in a control for disputes over merchandise trade, can be estimated for the medium and high bin. However, when we re-estimate Model 5 on the low bin quasi separation is caused on the disputes control variable, indicating that the control variable explains much of the variation in the outcome

for the low bin. The results of the model re-estimates for each bin type is shown in Table 4.

Across all models, the low bin exhibits no statistically significant relationship between compliance and number of third parties. In fact, within the low bin, it is not until Model 5 do the coefficients on number of third parties indicate a negative relationship. In the medium and high bins, each model consistently estimates a negative relationship between compliance and third parties but it is not significant. Both of these findings are consistent with the results from the regression splines. In conjunction with the results from the splines, this suggests that we should reject the null hypothesis that third parties in small numbers have a substantial effect on compliance. Perhaps more importantly though, the bins suggest that the effects observed in Kucik and Peritz (2021) are likely an artifact of observing very few cases of third party participation beyond just a few third parties.

Table 4: Re-estimating Model Results using Binning

	Probit (Binary)			Probit (Ordinal)			Probit (Binary 5)	
	Low	Medium	High	Low	Medium	High	Medium	High
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
(Intercept)	1.053	0.894	5.583+				−0.518	6.757
	(2.078)	(1.085)	(3.165)				(1.479)	(4.604)
lnthird	0.359	−1.086	−1.660	1.621	−0.726	−0.895	−1.055	−2.832
	(2.225)	(0.737)	(1.068)	(2.202)	(0.626)	(0.867)	(1.021)	(1.921)
trade_share_rctp	1.106	1.246	1.952+	0.648	0.974	1.624+	3.211*	1.657
	(2.580)	(0.871)	(1.087)	(2.879)	(0.772)	(0.853)	(1.432)	(1.620)
times_ruled	−0.778	0.019+	−0.001	−0.904+	0.010	0.002	0.019	−0.022
	(0.540)	(0.011)	(0.011)	(0.536)	(0.008)	(0.009)	(0.015)	(0.016)
gdp_share	1.161	−0.707	−2.694	1.052	−0.530	−1.543	−0.837	−1.224
	(1.882)	(0.605)	(1.750)	(1.879)	(0.512)	(1.054)	(0.871)	(1.473)
r_greatpower		0.646	0.763	1.109	0.298	0.802	1.753**	0.705
		(0.418)	(1.053)	(1.775)	(0.372)	(0.697)	(0.663)	(1.456)
article_xxii	3.455	0.416	1.026	4.046+	0.184	0.995+	0.000	−0.237
	(2.235)	(0.381)	(0.757)	(2.162)	(0.340)	(0.583)	(0.576)	(1.160)
remedy	−1.385	0.529	−1.368+	−2.308	0.690*	−0.821	0.354	−1.617
	(1.791)	(0.379)	(0.815)	(1.827)	(0.338)	(0.586)	(0.536)	(1.047)
clarity	−2.436	−0.582	−1.950*	−3.125	−0.237	−1.530*	0.236	−1.358
	(1.907)	(0.424)	(0.810)	(2.021)	(0.375)	(0.629)	(0.658)	(0.951)
disputedprod_lnttrade							0.054	0.386
							(0.088)	(0.238)
Num.Obs.	29	76	48	29	76	48	62	33
AIC	28.8	89.5	51.2	41.9	144.2	77.1	60.7	42.4

¹ + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Similarly, we can use the results from the splines to test Hypothesis 2, namely that there is not a substantial difference between the effect of no third parties and one third party on compliance. We do this by generate the predicted values of compliance for no third party participation ($\ln(0+1)$) and one third party participation (roughly $\ln(3)$) for each Model 1,2, and 5. For Model 1 the difference in predicted compliance between no third parties and one third party is -0.125 with a z-score of -2.46 which is statistically significant. Similar results are found in Model 2. However, Model 5 the estimated difference drops to just -0.057 and the result is not statistically significant. It is hard to say from these results whether not the null hypothesis should be rejected. The best that we can say is that there is a statistically significant decrease in compliance associated with a change from no third parties to just one third party. However, this relationship dwindles when additional controls for industry level employment are added.

For our final contribution, we posit a control for state capacity via a domestic economic stress indicator that is not theoretically accounted for in Kucik & Peritz (2021). Furthermore, our proposed control indicator supports the authors' overarching argument that the defendant's capacity affects its ability to comply with WTO rulings by implementing resolutions in its domestic policies. Although Kucik & Peritz (2021) account for other financial variables, such as trade shares or shared GDP between litigants, the authors fail to include an indicator of lack of fiscal capacity, thus limiting the political capacity of the respondent state. We offer this as an alternative explanation for why respondents cannot comply and simply do not remove policies in violation of WTO rules. To account for this potential omitted confounder, we use a dichotomous banking crisis measure as a proxy in our newly estimated model in Table 5.

Arguably, a banking crisis measure is an ideal indicator of domestic economic stress because it focuses on local fiscal capacity, reduces cash flow circulation within the local economy, and requires large-scale government intervention to resolve. Theoretically, if there is less capital in circulation, the ability of bureaucratic and public spending to increase

decreases under these constraints. In the context of the Kucik & Peritz (2021) argument, rising levels of domestic economic stress, stemming from a domestic banking crisis, limit a government’s ability to remove costly policies, thereby decreasing incentives to comply with the WTO ruling. This is especially reasonable if the disputed policy gives the respondent country a significant economic advantage, such as enabling it to run a beneficial trade deficit and/or shielding local businesses from external competition in the local market. For this reason, we find it necessary to include domestic economic stress control in our extension.

Kucik & Peritz do not include any variables in capturing the true domestic economic conditions of a respondent state. The closest proxy variable supporting Kucik and Peritz (2021) ’s argument is the relative trade share between the respondent and the third-party participant. This measure captures the shared trade at stake in the dispute and, more intuitively, the greater this share is, the greater the incentives third parties have to opt into the dispute. While this variable provides valuable context, it does not extend to a broader story of economic strain and does not capture the underlying fiscal health of a respondent state’s ability to comply with domestic policy changes despite their potential willingness to do so. Especially, since this is a fair explanation that can be relevant to economies of all sizes: advanced, middle, and lower.

For the extension, we employ data from the Harvard Business School’s Behavioral Finance & Financial Stability (BFF) Global Crises dataset (Reinhart et al., 2016). The BFF dataset spans 1916 to 2016 and includes banking crises, systemic crises, currency crises, and inflation crises. The rationale for focusing on banking crises is to ensure that we capture truly local economic conditions. As mentioned previously, banking crises require significant government intervention and produce prolonged economic contractions, potentially depleting state capacity—and thereby reducing a state’s capacity to comply with a ruling, even if, under other conditions, compliance would be feasible or preferred. By contrast, inflation is often a byproduct of other monetary policy failures. A currency crisis indicator is also not considered in this project, as not all EU member states use the Euro as their official

national currency. Lastly, the BFFs dataset codes countries individually. To merge with Kucik and Peritz (2021) dataset, we recode the available EU member countries into a single EU. Although the BFF dataset covers the economic crisis history of 70 countries, it does not include 10 EU members. More specifically, despite these countries experiencing banking crises during this period, Bulgaria, Croatia, Cyprus, the Czech Republic, Estonia, Latvia, Lithuania, Luxembourg, Malta, Slovakia, and Slovenia are not included in the EU analysis. Theoretically, this is not a concern because the states that would truly topple the entire EU into a crisis are most likely to be among the leading economies, like France or Germany.

Finally, after re-running Kucik and Peritz (2021) models in addition to the binary (*econ_stress_b*) measure, we do not find substantially altered results (see Table 5). Overall, the core finding remains robust in significance and magnitude: third-party participation decreases compliance across all specifications. Furthermore, we observe a reduction in ρ , suggesting that although the original models carried minimal omitted-variable bias, our additional control slightly improves our overall model. Together, these results reaffirm the main findings of Kucik and Peritz (2021), remaining robust, after including and controlling for the domestic economic stress of the respective respondent states. Interestingly, we found a positive relationship between this control variable and compliance, perhaps offering marginal theoretical insights. Perhaps countries undergoing banking crises have greater incentives to comply with WTO dispute settlement to signal to the international community that they are open to trade and investment, as an alternative explanation to the compliance puzzle.

Table 5: Extension: Effect of Economic Stress (Banking Crisis) on WTO Compliance

	Compliance						Heckman (7)		Heckman (8)	
	Probit (1)	Ordered (2)	Linear (3)	Linear (4)	Probit (5)	Probit (6)	Ruling Selection	Compliance Outcome	Ruling Selection	Compliance Outcome
lnthird	−0.633** (0.197)	−0.534** (0.181)	−0.173** (0.052)	−0.316** (0.106)	−0.768** (0.271)	−0.830* (0.371)	0.632** (0.141)	−0.176* (0.069)	0.604** (0.140)	−0.140* (0.070)
trade_share_rctp	1.397* (0.584)	1.293* (0.571)	0.446** (0.167)	0.825* (0.339)	1.672** (0.619)	2.311* (1.164)	2.173** (0.412)	0.472* (0.207)	2.181** (0.391)	0.584** (0.212)
times_ruled	0.002 (0.009)	0.002 (0.007)	0.001 (0.002)	0.002 (0.005)	−0.004 (0.008)	−0.020 (0.015)	−0.003 (0.002)	0.001 (0.002)	0.003 (0.006)	0.001 (0.002)
gdp_share	−0.653 (0.411)	−0.444 (0.399)	−0.115 (0.113)	−0.227 (0.237)	−0.986* (0.405)	−0.494 (1.375)	−0.201 (0.269)	−0.081 (0.110)	−0.065 (0.262)	−0.089 (0.111)
r_greatpower	0.446 (0.360)	0.390 (0.309)	0.112 (0.106)	0.238 (0.206)	0.966* (0.484)		0.454* (0.223)	0.128 (0.093)	0.504* (0.228)	0.145 (0.094)
article_xxii	0.699* (0.282)	0.570* (0.251)	0.172* (0.080)	0.334* (0.160)	0.526 (0.400)	0.852 (0.663)	−0.294 (0.196)	0.123 (0.076)	−0.218 (0.195)	0.107 (0.078)
remedy	0.063 (0.302)	0.174 (0.285)	0.028 (0.085)	0.103 (0.174)	0.144 (0.362)	0.237 (0.997)	−0.193 (0.211)	0.088 (0.072)	−0.207 (0.186)	0.086 (0.072)
clarity	−0.806* (0.320)	−0.556 (0.310)	−0.212* (0.093)	−0.351 (0.192)	−0.810* (0.379)	−1.765 (1.025)				
econ_stress_b	0.580 (0.356)	0.586 (0.338)	0.129 (0.097)	0.308 (0.197)	0.387 (0.385)	0.371 (0.865)		0.156 (0.088)		0.152 (0.087)
disputedprod_lnttrade					0.086 (0.077)					
log(employ_share_tot)						−0.126 (0.178)				
systemic_new							0.751** (0.233)		0.677** (0.231)	
art_agric							−0.471 (0.289)			
art_sps							−0.765 (0.436)			
art_service							−0.935* (0.425)			
ratio_rule_yr									0.119 (0.953)	
adv_frequency									−0.700** (0.246)	
Constant	0.655 (0.714)		0.644** (0.202)	1.126** (0.408)	0.550 (0.629)	0.670 (2.153)	−1.611** (0.306)	0.534 (0.285)	−1.699** (0.440)	0.357 (0.290)
Observations	153	153	153	153	115	61	373		373	
R ²			0.231	0.207			0.201		0.205	
Adjusted R ²			0.182	0.157			0.146		0.150	
ρ							0.057		0.340	
Inverse Mills Ratio							0.022 (0.151)		0.136 (0.155)	

Note: $p < 0.1$; $*p < 0.05$; $**p < 0.01$.

Standard errors in parentheses. Models 1–6 use standard errors clustered by dispute group.

4 Conclusion

This paper has attempted to build upon the work of Kucik and Peritz (2021) and investigate the relationship between compliance WTO disputes and third party participation. Kucik and Peritz (2021)’s results that third party participation is correlated with a decrease in likelihood of respondent state compliance with WTO disputes is replicated and confirmed within this paper. However, key gaps in the theoretical design of their models and interpretation we’re attempted to be filled by the extensions in this paper.

First, on the one hand, while third party observers may increase compliance by putting eyes on the case and increasing the costs of non-compliance for respondents (Busch and Reinhardt, 2006), Kucik and Peritz (2021)’s argument argues that third parties may also obfuscate the ruling making compliance difficult even when states want to. To some extent, both arguments are about state capacity to respond in meaningful ways to rulings. Hence, this paper introduced a control for state capacity by introducing binary variable for banking crises in the year the ruling was made. While theoretically, this should explain whether states had the fiscal and overhead capacity to respond to WTO rulings, the findings did not substantially alter Kucik and Peritz (2021)’s results. However, the model with our control variable for state capacity did result in a better model fit, suggesting that we may be right to think of compliance as an argument about state capacity.

What our paper has shown is that in small numbers third participation has limited effects on compliance in all of Kucik and Peritz (2021)’s models. Regression splines and binning techniques illustrate that the substantial, and significant effects, of third party participation in WTO disputes does not really kick in till about six or seven third parties are involved. This is a worthy critique of Kucik and Peritz (2021)’s findings as very few cases in their dataset have above two or three third parties involved. Additionally, while there is a statistically significant effect of going from no third parties to one third party on compliance, the effect is not stable across all of Kucik and Peritz (2021)’s models and loses significance entirely after adding their additional controls. Lastly, while the binning technique does suffer from

arbitrary bin lengths and determination, the perfect and quasi-perfect separation results are interesting to note. Particularly, in the bin with no third parties involved separation occurs because 93% of the cases resulted in compliance. Furthermore, when that bin is expanded to include 0-2 third parties, roughly, separation is again caused by the great powers control variable—suggesting perhaps that in those instances the respondent state being a great power perfectly or quasi-perfectly predicts the outcome of non-compliance. However, these inferences cannot be borne out with standard errors on these results, which we do not investigate.

Broadly, all of our findings, null and otherwise, suggest that the story of third party participation in WTO disputes is not as simple as either eyes on the case or additional opinions in the courtroom. Future research should focus on the nuances of compliance with international law, including the marginal effects of third parties and state capacity might have on state incentives to comply with WTO rulings. Additionally, our findings should encourage other researchers who turn to linear models (generalized or otherwise) to investigate the deeper effects of their treatment variable on their outcome when their outcome is highly skewed.

References

- Busch, M. L. and Reinhardt, E. (2006). Three’s a Crowd: Third Parties and WTO Dispute Settlement. *World Politics*, 58(3):446–477.
- Chayes, A. and Chayes, A. H. (1993). On compliance. *International Organization*, 47(2):175–205.
- Downs, G. W., Rocke, D. M., and Barsoom, P. N. (1996). Is the Good News about Compliance Good News about Cooperation? *International Organization*, 50(3):379–406.
- Kucik, J. and Peritz, L. (2021). How Do Third Parties Affect Compliance in the Trade Regime? *The Journal of Politics*, 83(3):1184–1189.
- Maggi, G. (1999). The Role of Multilateral Institutions in International Trade Cooperation. *American Economic Review*, 89(1):190–214.
- Reinhart, C., Rogoff, K., Trebesch, C., and Reinhart, V. (2016). Global Crises Data by Country - Behavioral Finance & Financial Stability - Harvard Business School.

A Code Appendix

A.1 Replication

```
1 ### loading required packages
2 library(haven)
3 library(zeligverse)
4 library(sandwich)
5 library(lmtest)
6 library(sampleSelection)
7 library(stargazer)
8 library(MASS)
```

```

9 library(ordinal)
10 library(modelsummary)
11 library(clubSandwich)
12 library(patchwork)
13 library(ggplot2)
14
15 get_mode <- function(v) {
16   uniqv <- unique(v)
17   uniqv[which.max(tabulate(match(v, uniqv)))]
18 }
19
20 # load their data set
21 data <- read_dta("dataverse_files (6)/Kucik_Peritz_JOP2020.dta")
22
23 third <- data$thirdparties
24 zeros <- sum(data$thirdparties == 0, na.rm=T)
25
26
27 # summary stadata# summary stats and visuals
28 # restricts the data to those cases that had a ruling
29 data.subset <- subset(data, comply_b ==1 | comply_b == 0, select =
30                       c("ds", "comp1", "resp1",
31                         "comply_b", "comply_n", "lnthird", "thirdparties
32                         ",
33                         "trade_share_rctp", "gdp_share", "r_greatpower",
34                         "times_ruled", "lnclaims", "article_xxii", "
35                         remedy", "clarity",
36                         "dispute_combined"))
37
38
39 third_parties <- data.subset$thirdparties
40 zeros <- sum(data.subset$thirdparties == 0, na.rm=T)
41
42
43 par(mfrow = c(1, 2))

```



```

40 hist(third,
41       xlab = "Third Parties",
42       xlim = c(0,25),
43       main = "Histogram of Third Parties in All WTO Disputes in Dataset")
44 hist(third_parties,
45       xlab = "Third Parties",
46       xlim = c(0,25),
47       main = "Histogram of Third Parties in Cases Ending in a Ruling")
48
49 get_mode(data.subset$thirdparties)
50 mean(third_parties, na.rm = T)
51
52 # table 1: baseline estimates
53
54 # probit model (Model 1)
55 cluster_var <- data.subset$dispute_combined #clusters for standard errors
56
57 probit1 <- glm(
58   comply_b ~ lnthird + trade_share_rctp + times_ruled +
59   gdp_share + r_greatpower + article_xxii + remedy + clarity,
60   data = data.subset,
61   family = binomial(link = "probit")
62 )
63 probit1_cluster <- coeftest(probit1, vcov = vcovCL(probit1, cluster =
64   cluster_var))
65
66 # ordered probit model (Model 2)
67 probit2 <- clm(
68   as.factor(comply_n) ~ lnthird + trade_share_rctp + times_ruled +
69   gdp_share + r_greatpower + article_xxii + remedy + clarity,
70   data = data.subset,
71   link = "probit"

```

```

72 probit2.cluster <- coeftest(
73   probit2,
74   vcov = vcovCL(probit2, cluster = ~dispute_combined)
75 )
76
77 # linear regression models (Models 3 and 4)
78 lm1 <- lm(comply_b ~ lnthird + trade_share_rctp + times_ruled +
79           gdp_share + r_greatpower + article_xxii + remedy + clarity,
80           data = data.subset)
81 lm1_cluster <- coeftest(lm1, vcov=vcovCL(lm1, cluster = ~dispute_combined)
82 )
83
84 lm2 <- lm(comply_n ~ lnthird + trade_share_rctp + times_ruled +
85           gdp_share + r_greatpower + article_xxii + remedy + clarity,
86           data = data.subset)
87 lm2_cluster <- coeftest(lm2, vcov=vcovCL(lm2, cluster = ~dispute_combined)
88 )
89
90 # control models for bilateral imports (Model 5) and industry employment (
91   Model 6)
92
93 vars1 <- c("ds","comply_b","comply_n","lnthird","r_greatpower",
94           "trade_share_rctp","gdp_share","disputedprod_lntrade",
95           "times_ruled","lnclaims","article_xxii","remedy","clarity",
96           "dispute_combined")
97
98 data.subset2 <- na.omit(data[, vars1])
99
100 probit5 <- glm(
101   comply_b ~ lnthird + trade_share_rctp + times_ruled + gdp_share +
102   r_greatpower + article_xxii + remedy + clarity + disputedprod_lntrade,
103   data = data.subset2,
104   family = binomial(link = "probit")
105 )

```

```

102 probit5_cluster <- coeftest(probit5, vcov=vcovCL(probit5, cluster = ~
    dispute_combined))
103
104
105 vars2 <- c("ds","comply_b","comply_n","lnthird","r_greatpower",
106           "trade_share_rctp","gdp_share","employ_share_tot",
107           "times_ruled","lnclaims","article_xxii","remedy","clarity",
108           "dispute_combined")
109
110 data.subset3 <- data[complete.cases(data[, vars2]), vars2]
111
112 probit6 <- glm(
113   comply_b ~ lnthird + trade_share_rctp + times_ruled + gdp_share +
114   article_xxii + remedy + clarity + log(employ_share_tot),
115   data = data.subset3,
116   family = binomial(link = "probit")
117 )
118 probit6_cluster <- coeftest(probit6, vcov=vcovCL(probit6, cluster = ~
    dispute_combined))
119
120
121 ### Heckman Selection models
122 ## ruling selection
123 heck1 <- heckit(
124   selection = ruling ~ lnthird + trade_share_rctp + times_ruled +
    gdp_share +
125   r_greatpower + article_xxii + remedy + systemic_new +
126   art_agric + art_sps + art_service,
127
128   outcome = comply_b ~ lnthird + trade_share_rctp + times_ruled +
    gdp_share + r_greatpower + remedy + article_xxii,
129
130
131   data = data,

```

```

132   method = "ml"
133 )
134
135 coef_test(
136   heck1,
137   vcov = "CR2",
138   cluster = data$dispute_combined
139 )
140
141 ## compliance outcome
142
143 heck2 <- heckit(
144   selection = ruling ~ lnthird + trade_share_rctp + times_ruled +
      gdp_share +
145     r_greatpower + article_xxii + remedy + systemic_new +
146     ratio_rule_yr + adv_frequency,
147
148   outcome = comply_b ~ lnthird + trade_share_rctp + times_ruled +
      gdp_share + remedy + r_greatpower + article_xxii,
149
150
151   data = data,
152   method = "ml"
153 )
154
155
156 ### Print Table 1
157 se_list <- list(
158   sqrt(diag(vcovCL(probit1, cluster = ~dispute_combined))),
159   sqrt(diag(vcovCL(probit2, cluster = ~dispute_combined))),
160   sqrt(diag(vcovCL(lm1, cluster = ~dispute_combined))),
161   sqrt(diag(vcovCL(lm2, cluster = ~dispute_combined))),
162   sqrt(diag(vcovCL(probit5, cluster = ~dispute_combined))),
163   sqrt(diag(vcovCL(probit6, cluster = ~dispute_combined))),

```

```

164     NULL,
165     NULL
166 )
167
168 stargazer(
169     probit1, probit2, lm1, lm2, probit5, probit6, heck1, heck2,
170     se = se_list,
171     no.space = TRUE,
172     star.char = c(".", "*", "***"),
173     type = "latex"
174 )
175
176 ### Heckman selection equations
177 # outcome equation
178 stargazer(
179     heck1, heck2,
180     title = "Outcome Equation (Compliance)",
181     column.labels = c("Heck1", "Heck2"),
182     no.space = TRUE,
183     dep.var.labels.include = TRUE,
184     selection.equation = FALSE,
185     star.cutoffs = c(0.1, 0.05, 0.01),
186     star.char = c(".", "*", "***"),
187     type = "text"
188 )
189 # selection equation
190 stargazer(
191     heck1, heck2,
192     title = "Selection Equation (Ruling)",
193     column.labels = c("Heck1", "Heck2"),
194     no.space = TRUE,
195     dep.var.labels.include = TRUE,
196     selection.equation = TRUE,

```

```

197   star.cutoffs = c(0.1, 0.05, 0.01),
198   star.char = c(".", "*", "**"),
199   type = "text"
200 )

```

A.2 Extension - Splines and Bins

```

1 library(haven)
2 library(sandwich)
3 library(lmtest)
4 library(sampleSelection)
5 library(stargazer)
6 library(MASS)
7 library(ordinal)
8 library(modelsummary)
9 library(patchwork)
10 library(ggplot2)
11 library(clubSandwich)
12 library(detectseparation)
13 library(splines)
14 library(marginaleffects)
15 data <- read_dta("dataverse_files (6)/Kucik_Peritz_JOP2020.dta")
16
17 # restricts the data to those cases that had a ruling
18 data.subset <- subset(data, comply_b ==1 | comply_b == 0, select =
19                       c("ds", "comp1", "resp1",
20                         "comply_b", "comply_n", "lnthird",
21                         "trade_share_rctp", "gdp_share", "r_greatpower",
22                         "times_ruled", "lnclaims", "article_xxii",
23                         "remedy", "clarity",
24                         "dispute_combined"))

```

```

25 # for controls
26 vars1 <- c("ds","comply_b","comply_n","lnthird","r_greatpower",
27           "trade_share_rctp","gdp_share","disputedprod_lntrade",
28           "times_ruled","lnclaims","article_xxii","remedy","clarity",
29           "dispute_combined")
30 data.subset2 <- na.omit(data[, vars1])
31
32 vars2 <- c("ds","comply_b","comply_n","lnthird","r_greatpower",
33           "trade_share_rctp","gdp_share","employ_share_tot",
34           "times_ruled","lnclaims","article_xxii","remedy","clarity",
35           "dispute_combined")
36
37 data.subset3 <- data[complete.cases(data[, vars2]), vars2]
38
39 ###
40 ### Regression Splines
41 ###
42
43 probit_spline1 <- glm(
44   comply_b ~ ns(lnthird, df = 3) + trade_share_rctp + times_ruled +
45   gdp_share + r_greatpower + article_xxii + remedy + clarity,
46   data = data.subset,
47   family = binomial(link = "probit")
48 )
49
50 data.subset$comply_n <- as.factor(data.subset$comply_n)
51 probit_spline2 <- polr(
52   comply_n ~ ns(lnthird, df = 3) + trade_share_rctp + times_ruled +
53   gdp_share + r_greatpower + article_xxii + remedy + clarity,
54   data = data.subset,
55   method = "probit"
56 )
57

```

```

58 p1 <- plot_predictions(probit_spline1, condition = "lnthird")
59 p1 <- p1 + labs(title = "Marginal Predicted Effect of Third Parties on
      Compliance (Binary)",
60                x = "ln(third parties)")
61
62 p2 <- plot_predictions(probit_spline2, condition = "lnthird")+
63   facet_wrap(~group)
64 p2 <- p2 + labs(title = "Marginal Predicted Effect of Third Parties on
      Compliance (Ordinal)",
65                x = "ln(third parties)")
66
67 p1 + p2
68
69
70 probit_spline3 <- glm(
71   comply_b ~ ns(lnthird, df = 3) + trade_share_rctp + times_ruled +
      gdp_share +
72   r_greatpower + article_xxii + remedy + clarity + disputedprod_lntrade,
73   data = data.subset2,
74   family = binomial(link = "probit")
75 )
76
77 probit_spline4 <- glm(
78   comply_b ~ ns(lnthird, df = 3) + trade_share_rctp + times_ruled +
      gdp_share +
79   article_xxii + remedy + clarity + log(employ_share_tot),
80   data = data.subset3,
81   family = binomial(link = "probit")
82 )
83
84 p3 <- plot_predictions(probit_spline3, condition = "lnthird")+
85   labs(title = "Marginal Predicted Effect of Third Parties on Compliance (
      Binary)",

```



```

86     subtitle = "Controlling for Bilateral Disputes onM Merchandise
Imports",
87     x = "ln(third parties)")
88 p4 <- plot_predictions(probit_spline4, condition = "lnthird")+
89 labs(title = "Marginal Predicted Effect of Third Parties on Compliance (
Binary)",
90     subtitle = "Controlling for Industry Employment Share",
91     x = "ln(third parties)")
92
93 p3 +p4
94
95 comparisons(
96     probit_spline1,
97     variables = list(lnthird = c(0, log(3))),
98     newdata = datagrid(lnthird = 0),
99 )
100
101
102 predictions(
103     probit_spline1,
104     newdata = datagrid(lnthird = c(0, log(2), log(3), log(4), log(5), log(6)
, log(7)))
105 )
106
107
108
109 ###
110 ### Bin into number of third parties (lnthird)
111 ###
112
113 lnthird <- data.subset$lnthird
114 hist(lnthird)
115

```

```

116 # Bin 1 (lnthird = 0 - 1)
117 data_low <- subset(data.subset, lnthird >= 0 & lnthird < 1.0)
118 data_low5 <- subset(data.subset2, lnthird >= 0 & lnthird < 1.0)
119
120 probit_low1 <- glm(
121   comply_b ~ lnthird + trade_share_rctp + times_ruled +
122     gdp_share + r_greatpower + article_xxii + remedy + clarity,
123   data = data_low,
124   family = binomial(link = "probit")
125 )
126 # analyze the separation
127 table(data_low$comply_b)
128
129 data_low[data_low$comply_b == 0, ]
130
131 aggregate(. ~ comply_b, data = data_low[, c("comply_b", "lnthird", "
    trade_share_rctp",
132                                           "times_ruled", "gdp_share", "
    r_greatpower",
133                                           "article_xxii", "remedy", "
    clarity")],
134           FUN = mean)
135
136 # drop great powers
137 probit_low1 <- glm(
138   comply_b ~ lnthird + trade_share_rctp + times_ruled +
139     gdp_share + article_xxii + remedy + clarity,
140   data = data_low,
141   family = binomial(link = "probit")
142 )
143
144 probit_low2 <- glm(
145   as.factor(comply_n) ~ lnthird + trade_share_rctp + times_ruled +

```

```

146     gdp_share + r_greatpower + article_xxii + remedy + clarity,
147     data = data_low,
148     link = "probit"
149 )
150
151 probit_low5 <- glm(
152     comply_b ~ lnthird + trade_share_rctp + times_ruled + gdp_share
153     + article_xxii + remedy + clarity + disputedprod_lnttrade,
154     data = data_low5,
155     family = binomial(link = "probit")
156 )
157
158 data_low5[data_low5$comply_b == 0, ]
159
160 aggregate(. ~ comply_b, data = data_low5[, c("comply_b", "lnthird", "
      trade_share_rctp",
161                                           "times_ruled", "gdp_share", "
      r_greatpower",
162                                           "article_xxii", "remedy", "
      clarity","disputedprod_lnttrade")],
163           FUN = mean)
164 # drop great powers and clarity and others DONT USE
165 probit_low5 <- glm(
166     comply_b ~ lnthird + trade_share_rctp + gdp_share
167     + disputedprod_lnttrade,
168     data = data_low5,
169     family = binomial(link = "probit")
170 )
171
172
173 # Bin 3 (data_low# Bin 3 (lnthird = 1 - 2)
174 data_medium <- subset(data.subset, lnthird >= 1.0 & lnthird < 2.0)
175 data_medium5 <- subset(data.subset2, lnthird >= 1.0 & lnthird < 2.0)

```

```

176
177 table(data_medium$comply_b)
178
179
180 probit_medium1 <- glm(
181   comply_b ~ lnthird + trade_share_rctp + times_ruled +
182     gdp_share + r_greatpower + article_xxii + remedy + clarity,
183   data = data_medium,
184   family = binomial(link = "probit")
185 )
186
187 probit_medium2 <- clm(
188   as.factor(comply_n) ~ lnthird + trade_share_rctp + times_ruled +
189     gdp_share + r_greatpower + article_xxii + remedy + clarity,
190   data = data_medium,
191   link = "probit"
192 )
193
194 probit_medium5 <- glm(
195   comply_b ~ lnthird + trade_share_rctp + times_ruled + gdp_share +
196     r_greatpower + article_xxii + remedy + clarity + disputedprod_lntrade,
197   data = data_medium5,
198   family = binomial(link = "probit")
199 )
200
201 # Bin 4 (lnthird = 2 - )
202 data_high <- subset(data.subset, lnthird >= 2.0 )
203 data_high5 <- subset(data.subset2, lnthird >= 2.0 )
204
205 table(data_high$comply_b)
206
207 probit_high1 <- glm(
208   comply_b ~ lnthird + trade_share_rctp + times_ruled +

```

```

209     gdp_share + r_greatpower + article_xxii + remedy + clarity,
210     data = data_high,
211     family = binomial(link = "probit")
212 )
213
214 probit_high2 <- clm(
215     as.factor(comply_n) ~ lnthird + trade_share_rctp + times_ruled +
216     gdp_share + r_greatpower + article_xxii + remedy + clarity,
217     data = data_high,
218     link = "probit"
219 )
220
221 probit_high5 <- glm(
222     comply_b ~ lnthird + trade_share_rctp + times_ruled + gdp_share +
223     r_greatpower + article_xxii + remedy + clarity + disputedprod_lntrade,
224     data = data_high5,
225     family = binomial(link = "probit")
226 )
227
228 ###
229 ### Print Model Summary
230 ###
231 modelsummary(
232     list(
233         "Probit (Binary) - Low"      = probit_low1,
234         "Probit (Binary) - Medium" = probit_medium1,
235         "Probit (Binary) - High"    = probit_high1
236     ),
237     stars = TRUE,
238     gof_map = c("nobs", "aic"),
239     vcov = list(
240         data_low$dispute_combinded,
241         data_medium$dispute_combinded,

```

```

242     data_high$dispute_combinded
243   ),
244   output = "latex"
245 )
246
247 modelsummary(
248   list(
249     "Probit (Ordinal) - Low"      = probit_low2,
250     "Probit (Ordinal) - Medium" = probit_medium2,
251     "Probit (Ordinal) - High"    = probit_high2
252   ),
253   stars = TRUE,
254   gof_map = c("nobs", "aic"),
255   vcov = list(
256     data_low$dispute_combinded,
257     data_medium$dispute_combinded,
258     data_high$dispute_combinded
259   ),
260   coef_omit = "\\| "
261 )
262
263 modelsummary(
264   list(
265     "Probit (Binary) 5 - Medium" = probit_medium5,
266     "Probit (Binary) 5 - High"   = probit_high5
267   ),
268   stars = TRUE,
269   gof_map = c("nobs", "aic"),
270   vcov = list(
271     data_medium5$dispute_combinded,
272     data_high5$dispute_combinded
273   ),
274   output = "latex"

```

A.3 Banking Crisis

```

1
2 # BANKING CRISES CONFOUNDER (H3) + DATA MERGING
3 #libraries:
4 library(haven)
5 library(dplyr)
6 library(ggplot2)
7 library(maps)
8 library(sandwich)
9 library(lmtest)
10 library(sampleSelection)
11 library(stargazer)
12 library(MASS)
13 library(openxlsx)
14 library(xtable)
15
16 #load in data
17 data <- read_dta("/Users/jessicagallardo/Dropbox/Stats II Replication/
    Kucik_Peritz_JOP2020.dta")
18
19 #load in econ data- BFFs Project
20 bffs_data <- read.xlsx("/Users/jessicagallardo/Dropbox/Stats II
    Replication/20160923_global_crisis_data.xlsx")
21
22 #clean data: keeping only select years
23 unique(data$year)
24 bffs_clean <- filter(bffs_data, Year >= 1995 & Year <= 2010)
25
26 #making a EU member list

```

```

27 EU_member <- c("Czechia", "Hungary", "Poland", "Romania", "Sweden",
28               "Austria", "Belgium", "Bulgaria", "Croatia", "Cyprus", "
    Estonia",
29               "Finland", "France", "Germany", "Greece", "Ireland", "Italy
    ",
30               "Latvia", "Lithuania", "Luxembourg", "Malta", "Netherlands
    ", "Portugal",
31               "Slovakia", "Slovenia", "Spain")
32
33 #respondent members
34 respondents <- c("United States", "Japan", "EU", "Australia", "Canada", "
    Turkey",
35               "Brazil", "India", "Indonesia", "Argentina", "Korea", "
    Chile",
36               "Mexico", "Guatemala", "Egypt", "Dominican Republic", "
    China",
37               "Colombia", "Thailand", "Malaysia", "Poland", "Venezuela
    ",
38               "Hungary", "Pakistan", "Philippines", "Peru", "Greece",
39               "Netherlands", "Slovak Republic", "Czech Republic", "
    South Africa",
40               "Ecuador", "Trinidad and Tobago", "Nicaragua", "Romania",
41               "Uruguay", "Croatia", "Panama", "Armenia")
42
43 #filtering BFFs dataset for relevant countries
44 combined <- unique(c(EU_member, respondents))
45 bffs_clean <- filter(bffs_clean, Country %in% combined)
46
47 #creating new EU aggregate (time dependent of EU states)
48 eu_join <- data.frame(
49   Country = EU_member,
50   join_year = c(2004, 2004, 2004, 2007, 1995,
51               1995, 1958, 2007, 2013, 2004, 2004,

```



```

52         1995, 1958, 1958, 1981, 1973, 1958,
53         2004, 2004, 1958, 2004, 1958,
54         1986, 2004, 2004, 1986)
55 )
56
57 bffs_clean <- bffs_clean |>
58   left_join(eu_join, by = "Country")
59
60 #adding up crises for the respective year w/ official EU members accounted
61 bffs_clean$Banking.Crisis <- as.numeric(bffs_clean$Banking.Crisis)
62 eu_banking_crises_sum <- bffs_clean |>
63   filter(!is.na(join_year)) |>
64   filter(Year >= join_year) |>
65   group_by(Year) |>
66   summarise(
67     eu_banking_crises_sum = sum(Banking.Crisis, na.rm = T),
68     eu_n_members = n()) |>
69   ungroup() |>
70   mutate(econ_stress_b = ifelse(eu_banking_crises_sum >= 3, 1, 0))
71
72 ##Merging Datasets onto full data object!!
73
74 # Step 1: merge EU crisis data onto full data
75 data <- data %>%
76   left_join(
77     eu_banking_crises_sum %>% dplyr::select(Year, econ_stress_b),
78     by = c("syear" = "Year")
79   ) %>%
80   mutate(econ_stress_b = ifelse(resp1 == "EU", econ_stress_b, NA))
81
82 # Step 2: merge non-EU country crisis data onto full data
83 data <- data %>%
84   mutate(resp1_bffs = dplyr::recode(resp1,

```

```

85         "USA" = "United States",
86         "Slovak Republic" = "Slovakia")) %>%
87   left_join(
88     bffs_clean %>% dplyr::select(Country, Year, Banking.Crisis),
89     by = c("resp1_bffs" = "Country", "syear" = "Year")
90   ) %>%
91   mutate(econ_stress_b = ifelse(is.na(econ_stress_b), Banking.Crisis,
92     econ_stress_b)) %>%
93   dplyr::select(-Banking.Crisis, -resp1_bffs)
94
95 #sanity checks
96 colnames(data)
97 sum(is.na(data$econ_stress_b))
98
99 #NEW extended df - this is for Heckman models (5 & 6)
100 ext_data <- data
101
102 #subsetting for only adverse rulings for Models 1-4
103 ext_data_sub <- subset(data, comply_b == 1 | comply_b == 0)
104
105 #write.csv(ext_data, "/Users/jessicagallardo/Dropbox/Stats II Replication/
106   ext_data.csv", row.names = F)
107
108
109
110
111
112
113
114 #LOAD IN NEW DATA - in case people want to start with this and not
115   recreate it

```

```

115 ext_data <- read.csv("/Users/jessicagallardo/Dropbox/Stats II Replication/
    ext_data.csv")
116
117 # create the subset for models 1-6
118 #binary compliance DV!!
119 ext_data_sub <- subset(ext_data, comply_b == 1 | comply_b == 0)
120
121 #Actual Extension:
122 # MODEL 1: Probit (binary DV)
123 m1_ext <- glm(comply_b ~ lnthird + trade_share_rctp + times_ruled +
124               gdp_share + r_greatpower + article_xxii + remedy +
125               clarity + econ_stress_b,
126               data = ext_data_sub,
127               family = binomial(link = "probit"))
128 m1_ext.cluster <- coeftest(m1_ext,
129                             vcov = vcovCL(m1_ext,
130                                             factor(
131                             ext_data_sub$dispute_combined)))
132
133 # MODEL 2: Ordered Probit (ordinal DV)
134 m2_ext <- polr(as.factor(comply_n) ~ lnthird + trade_share_rctp +
135               times_ruled +
136               gdp_share + r_greatpower + article_xxii + remedy +
137               clarity + econ_stress_b,
138               data = ext_data_sub,
139               method = "probit")
140 m2_ext.cluster <- coeftest(m2_ext,
141                             vcov = vcovCL(m2_ext,
142                                             factor(
143                             ext_data_sub$dispute_combined)))
144
145 # MODEL 3: OLS (binary DV)

```

```

144 m3_ext <- lm(comply_b ~ lnthird + trade_share_rctp + times_ruled +
145               gdp_share + r_greatpower + article_xxii + remedy +
146               clarity + econ_stress_b,
147               data = ext_data_sub)
148 m3_ext.cluster <- coeftest(m3_ext,
149                             vcov = vcovCL(m3_ext,
150                                             factor(
151                             ext_data_sub$dispute_combined)))
152 # MODEL 4: OLS (ordinal DV)
153 m4_ext <- lm(comply_n ~ lnthird + trade_share_rctp + times_ruled +
154               gdp_share + r_greatpower + article_xxii + remedy +
155               clarity + econ_stress_b,
156               data = ext_data_sub)
157 m4_ext.cluster <- coeftest(m4_ext,
158                             vcov = vcovCL(m4_ext,
159                                             factor(
160                             ext_data_sub$dispute_combined)))
161 # MODEL 5
162 m5_ext <- glm(comply_b ~ lnthird + trade_share_rctp + times_ruled +
163               gdp_share + r_greatpower + article_xxii + remedy +
164               clarity + econ_stress_b + disputedprod_lntrade,
165               data = ext_data_sub,
166               family = binomial(link = "probit"))
167
168 m5_ext.cluster <- coeftest(m5_ext, vcov =vcovCL(m5_ext,
169                                             factor(
170                             ext_data_sub$dispute_combined)))
171
172
173

```

```

174 ## Control for industry employment share (US and EU respondents only)
175 ext_data_sub_for_m6_ext <- na.omit(subset(ext_data, select = c("ds", "
      comply_b", "comply_n",
176                                     "lnthird", "
      r_greatpower",
177                                     "trade_share_rctp", "
      gdp_share",
178                                     "employ_share_tot", "
      times_ruled",
179                                     "article_xxii", "
      remedy", "clarity",
180                                     "econ_stress_b", "
      dispute_combined"))))
181 # MODEL 6
182 m6_ext <- glm(comply_b ~ lnthird + trade_share_rctp + times_ruled +
183               gdp_share + r_greatpower + article_xxii + remedy +
184               clarity + econ_stress_b + log(employ_share_tot),
185               data = ext_data_sub_for_m6_ext,
186               family = binomial(link = "probit"))
187
188 m6_ext.cluster <- coeftest(m6_ext, vcov = vcovCL(m6_ext,
189                                                  factor(
      ext_data_sub_for_m6_ext$dispute_combined)))
190
191 # MODEL 7: Heckman Selection
192 m7_ext <- heckit(
193   selection = ruling ~ lnthird + trade_share_rctp + times_ruled +
194     gdp_share + r_greatpower + article_xxii + remedy +
195     systemic_new + art_agric + art_sps + art_service,
196   outcome = comply_b ~ lnthird + trade_share_rctp + times_ruled +
197     gdp_share + r_greatpower + remedy + article_xxii + econ_stress_b,
198   data = ext_data, method = "2step"
199 )

```

```

200
201 # MODEL 8: Heckman Selection (alternative)
202 m8_ext <- heckit(
203   selection = ruling ~ lnthird + trade_share_rctp + times_ruled +
204     gdp_share + r_greatpower + article_xxii + remedy +
205     systemic_new + ratio_rule_yr + adv_frequency,
206   outcome = comply_b ~ lnthird + trade_share_rctp + times_ruled +
207     gdp_share + r_greatpower + remedy + article_xxii + econ_stress_b,
208   data = ext_data, method = "2step"
209 )
210
211 # Print results
212 stargazer(m1_ext.cluster, type = "text")
213 stargazer(m2_ext.cluster, type = "text")
214 stargazer(m3_ext.cluster, type = "text")
215 stargazer(m4_ext.cluster, type = "text")
216 stargazer(m5_ext.cluster, type = "text")
217 stargazer(m6_ext.cluster, type = "text")
218 stargazer(m7_ext, type = "text")
219 stargazer(m8_ext, type = "text")
220
221
222
223 #GENERATING LATEX TABLES
224 # Table 1: Base models
225 stargazer(m1_ext.cluster, m2_ext.cluster, m3_ext.cluster, m4_ext.cluster,
226           no.space = TRUE,
227           star.cutoffs = c(0.1, 0.05, 0.01),
228           star.char = c(".", "*", "**"),
229           type = "latex")
230
231 # Table 2: Robustness models + Heckman
232 stargazer(m5_ext.cluster, m6_ext.cluster, m7_ext, m8_ext,

```

```

233     no.space = TRUE,
234     star.cutoffs = c(0.1, 0.05, 0.01),
235     star.char = c(".", "*", "**"),
236     type = "latex")
237
238
239 #fixing formatting for Heckman models
240 summary(m7_ext)
241 summary(m8_ext)

```

A.4 Descriptives: Table 1 and Figure 1

```

1
2 # ADDITIONAL TABLES & FIGURES
3 #Figures FOR PAPER
4
5 #sanity check for banking crises (Not in paper)
6 ext_data_sub %>%
7   dplyr::select(resp1, syear, econ_stress_b) %>%
8   distinct() %>%
9   ggplot(aes(x = syear, y = resp1, fill = factor(econ_stress_b))) +
10  geom_tile(color = "white") +
11  scale_fill_manual(values = c("0" = "grey80", "1" = "black"),
12                    labels = c("No Banking Crisis", "Banking Crisis"),
13                    name = "Economic Stress",
14                    na.value = "white") +
15  labs(title = "Banking Crises Among WTO Respondents (1995-2010)",
16        x = "Year", y = "Country") +
17  theme_light() +
18  theme(axis.text.y = element_text(size = 8))
19
20

```

```

21 ext_data_sub %>%
22   group_by(resp1) %>%
23   summarise(
24     total_disputes = n(),
25     n_comply = sum(comply_b, na.rm = TRUE),
26     n_noncomply = sum(comply_b == 0, na.rm = TRUE),
27     compliance_rate = mean(comply_b, na.rm = TRUE)
28   ) %>%
29   arrange(desc(total_disputes)) %>%
30   print(n = 30)
31
32
33
34
35 #official table #1 in paper
36 dyad_table <- ext_data_sub %>%
37   mutate(
38     country_a = pmin(comp1, resp1),
39     country_b = pmax(comp1, resp1)
40   ) %>%
41   group_by(country_a, country_b) %>%
42   summarise(
43     total_disputes = n(),
44     n_comply = sum(comply_b, na.rm = TRUE),
45     n_noncomply = sum(comply_b == 0, na.rm = TRUE),
46     compliance_rate = mean(comply_b, na.rm = TRUE),
47     .groups = "drop"
48   ) %>%
49   filter(total_disputes >= 2) %>%
50   arrange(desc(total_disputes))
51
52 print(dyad_table, n = 30)
53

```



```

54
55
56
57 #Table 1 in Latex
58 dyad_table_clean <- dyad_table %>%
59   rename(
60     'Country A' = country_a,
61     'Country B' = country_b,
62     'Total Disputes' = total_disputes,
63     'N Comply' = n_comply,
64     'N Non-Comply' = n_noncomply,
65     'Compliance Rate' = compliance_rate
66   ) %>%
67   mutate('Compliance Rate' = round('Compliance Rate', 2))
68
69 print(xtable(dyad_table_clean,
70             caption = "Repeated Dispute Dyads and Compliance in WTO
71             Adverse Rulings",
72             label = "tab:dyads"),
73       include.rownames = FALSE,
74       caption.placement = "top")
75
76
77 #FIGURE 1: Mapping Frequency of WTO Adverse Rulings
78
79 # Counting ONLY adverse rulings by respondent country
80 ruling_counts <- data %>%
81   filter(comply_b == 1 | comply_b == 0) %>%
82   group_by(resp1) %>%
83   summarise(times_sued = n()) %>%
84   mutate(resp1 = ifelse(resp1 == "Slovak Republic", "Slovakia", resp1))
85

```

```

86 # creating a vector of EU member states for mapping
87 eu_countries <- c("Austria", "Belgium", "Bulgaria", "Croatia", "Cyprus", "
    Czech Republic",
88                  "Denmark", "Estonia", "Finland", "France", "Germany", "
    Greece",
89                  "Hungary", "Ireland", "Italy", "Latvia", "Lithuania", "
    Luxembourg",
90                  "Malta", "Netherlands", "Poland", "Portugal", "Romania",
    "Slovakia",
91                  "Slovenia", "Spain", "Sweden", "United Kingdom")
92
93 # Getting EU count
94 eu_count <- ruling_counts %>% filter(resp1 == "EU") %>% pull(times_sued)
95
96 # Expand EU to individual countries
97 ruling_counts_expanded <- ruling_counts %>%
98   filter(resp1 != "EU") %>%
99   bind_rows(data.frame(resp1 = eu_countries, times_sued = eu_count)) %>%
100   filter(resp1 != "Canada")
101
102 # Getting world map data
103 world <- map_data("world")
104
105 # Joining data
106 map_data_joined <- world %>%
107   left_join(ruling_counts_expanded, by = c("region" = "resp1"))
108
109 # Plotting map
110 ggplot(map_data_joined, aes(x = long, y = lat, group = group, fill =
    times_sued)) +
111   geom_polygon(color = "white", linewidth = 0.1) +
112   scale_fill_distiller(palette = "Blues", direction = 1,
113                        na.value = "grey80",

```

```

114         name = "Times Sued",
115         breaks = c(10, 20, 30, 40, 50),
116         limits = c(1, 58)) +
117 labs(title = "Frequency of WTO Adverse Rulings by Defendant Country
        (1995-2012)",
118       subtitle = "155 Total Adverse Rulings") +
119 theme_void() +
120 theme(legend.position = "bottom",
121       plot.title = element_text(hjust = 0.5),
122       plot.subtitle = element_text(hjust = 0.5),
123       plot.caption = element_text(hjust = 0.5, size = 11))

```